

Counterfactual Simulation of Extreme Mental Health Scenarios for Clinical Preparedness via Fine-Tuned LLMs and Explainable AI

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Abstract—Mental health crisis episodes such as suicide attempts, psychotic breaks, and substance induced decompensation will tend to have warning signs in the clinical notes days or weeks before, but clinicians do not have right systematic tools to predict such possible negative trajectories. Existing AI systems are binary risk classifiers, which do not answer the crucial question: What would that trajectory be like if this patient do worse? The purpose of this research was to create a counterfactual reasoning model which is explainable and generates possible extreme mental health situations based on clinical evidence. We presented a multi-stage pipeline that includes a fine-tuned large language model to extract clinical factors, detect negation-aware safety signals, generate evidence-based scenarios with Tree-of-Thoughts reasoning, and attribute evidence back to source text with post-hoc. The model was trained with 94,592 MIMIC-IV discharge summaries, with a cross-entropy loss of 1.706 and a token level accuracy of 67.5%. The qualitative analysis showed that the system is effective to generate clinically coherent scenarios and find nexus factors evidence of multiple crisis pathways. This framework provides clinicians with a preparedness oriented framework that goes beyond risk scores to actionable scenario planning, which may allow intervention in psychiatric care earlier. The Codebase and implementation is made available at <https://github.com/lakshmidhatrigangavarapu/CFSP> for reproducibility.

Keywords—Counterfactual Reasoning, Large Language Model, Clinical NLP, Explainable AI, LoRA Fine-Tuning, Scenario Simulation, DSM-5, Tree Of Thoughts.

I. INTRODUCTION

Mental health problems are the greatest cause of years lived with disability in the world, with depression alone impacting around 280 million people [1]. Psychiatric encounters especially in the clinical documentation produced during the encounter especially in the discharge summaries have rich longitudinal data in the form of symptom patterns, medication history, social stressors and crisis events. But this unstructured text data is still systematically underutilized in proactive clinical reasoning [2]. Although recent developments in natural language processing and large language models have

enabled better automated analysis of clinical text [3] [4], they have mostly been applied to classification tasks as opposed to mental health risk assessment.

Current AI mental health applications are binary risk classifications (high/medium/low), instead of answering the most crucial clinical question: *If this patient worsens, what would that course of action be like, and how can we prepare?* The present methods have three limitations. Firstly to begin with, risk stratification systems reduce the large space of patient trajectories to a single categorical label or probability score, giving no actionable information on what adverse trajectory may occur [13]. Second, neural classifiers function as black boxes clinicians cannot verify whether predictions are grounded in documented evidence or reflect spurious correlations [10]. Third, current systems are reactive rather than preparatory: they alert clinicians *that* risk exists but provide no structured reasoning about *what* to prepare for. No existing system generates structured narratives of potential adverse trajectories with traceable evidence attribution. This gap is particularly critical in mental health, where crisis events such as suicide attempts, psychotic breaks, and substance-induced decompensation often exhibit documented warning signs days or weeks prior [2] [6].

In this paper, we address this gap by proposing an explainable counterfactual reasoning framework that, rather than predicting outcomes, generates structured narratives of plausible crisis trajectories grounded in clinical evidence. The framework operates as a clinical decision support tool through a multi-stage pipeline with five key components:

- 1) **Clinical Factor Extraction:** A LoRA fine-tuned Phi-3.5-Mini-Instruct model extracts structured clinical factors from discharge summaries using a DSM-5/ICD-11 grounded schema, with parameter-efficient adaptation of only 1.30% of total parameters trained on 94,592 MIMIC-IV records.
- 2) **Post-Extraction Validation:** ICD code normalization,

context enrichment for housing, trauma, and functional decline indicators, and a negation-aware safety signal detection algorithm that distinguishes active threats (e.g., “threatened to jump”) from documented denials (e.g., “denies suicidal ideation”) using a contextual lookback window.

- 3) **Evidence-Based Branch Gating:** A gating mechanism that conditionally activates each of three scenario pathways (psychiatric deterioration, substance escalation, social collapse) only when supporting clinical evidence exists, preventing hallucinated scenarios.
- 4) **Tree-of-Thoughts Scenario Generation:** This is a framework of reasoning that generates counterfactual crisis scenarios on active branches sequentially by identifying triggers, reasoning causally, and synthesizing scenarios.
- 5) **Evidence Attribution with Nexus Detection:** A post-hoc explainability layer that traces each generated claim to particular source text spans and finds nexus factors evidence supporting many different crisis pathways in parallel to prioritize targets of intervention.

This paper adds the initial end-to-end system that formalized clinical extraction, safety-conscious validation, constrained scenario generation, and attributable evidence of mental health crisis preparedness. This framework aims to assist clinicians, psychiatrists, and healthcare systems, making it possible to proactively prepare and plan an intervention in crises, support decisions with scenarios, and psychiatric risk assessments, hospital discharge planning, and real-time clinical monitoring. It should be stressed that the suggested system is not supposed to generate deterministic clinical predictions but, instead, it produces counterfactual, evidence-based situations that assist clinicians in their reasoning about potential counterfactual adverse progressions and preparedness measures.

II. RELATED WORK

NLP on clinical text has shown good performance on mental health tasks. Teferra et al. [2] demonstrated that the clinical notes text features add greater value to risk prediction compared to structured EHR markers. Sezgin et al. [3] showed that it was possible to extract clinical entities in patient generated health data. Lho et al. [13] obtained an AUROC of over 0.90 in the detection of depression when using the embeddings of LLM. Nonetheless, the current methods all use clinical notes as inputs for classification. None of them produce systematic accounts of possible negative paths. Our labor is different: we extract as a preparation to reason, but not as a goal. Within the context of large language models in mental health, Xu et al. [4] proposed Mental-LLM, showing that domain-specific fine-tuning is a necessary condition to achieve reliable mental health prediction. Hua et al. [6] and Guo et al. [5] found that there were enduring gaps in safety assessment and explainability. DSM-5 grounded frameworks were examined by Kim and Wang [14]. Regardless of these developments, none of the existing systems produce structured

extreme scenario narratives with attributable evidence of their traces.

Counterfactual reasoning has been receiving more and more attention in healthcare AI. Prosperi et al. [8] defined the clinical potential of counterfactual AI and asked the question: What would happen in case condition X were present? Gat et al. [9] showed that fine-tuned LLMs are capable of producing plausible counterfactuals, whereas Chen et al. [15] discovered that general-purpose LLMs cannot reason about high stakes in clinical settings. Our model realizes counterfactual simulation in the case of extreme mental health paths.

In terms of explainability in clinical AI, classical XAI methods like SHAP, LIME, and attention visualization have been criticized to be lacking in clinical actionability [10] a clinician cannot take action on feature X had importance 0.23. Another alternative is the emerging Generative XAI that interprets model reasoning as clinical narratives [11]. Our evidence attribution offers the post-hoc interpretability by basing outputs on particular input spans, which allows clinicians to check the plausibility of scenarios with source documentation.

III. PROBLEM FORMULATION

A. Formal Definition

Let $\mathcal{N}$ denote a discharge summary (Clinical Note) represented as a character sequence $\mathcal{N} = (c_1, c_2, \dots, c_n)$. Let $\mathcal{S}$ denote a structured extraction schema with k fields:

$$\mathcal{S} = \{f_1, f_2, \dots, f_k\} \quad \text{where } f_i \in \mathcal{V}_i \quad (1)$$

where $\mathcal{V}_i$ is the vocabulary/value space for field i (e.g., ICD codes, severity levels, boolean indicators).

Our objective is to learn a composite function:

$$\mathcal{F} : \mathcal{N} \rightarrow (\mathcal{S}, \mathcal{R}, \mathcal{E}) \quad (2)$$

where:

- $\mathcal{S}$: Structured clinical factors (14-field DSM-5/ICD-11 schema)
- $\mathcal{R}$: Set of scenario narratives $\{R_A, R_B, R_C\}$ for three branches
- $\mathcal{E}$: Evidence attribution mapping $e : \mathcal{R} \rightarrow 2^{\mathcal{N}}$

B. Pipeline Decomposition

The function $\mathcal{F}$ decomposes into seven stages:

$$\mathcal{F}_1 : \mathcal{N} \rightarrow \mathcal{S}_{\text{raw}} \quad (\text{Extraction}) \quad (3)$$

$$\mathcal{F}_2 : \mathcal{S}_{\text{raw}} \times \mathcal{N} \rightarrow \mathcal{S}_{\text{norm}} \quad (\text{Normalization}) \quad (4)$$

$$\mathcal{F}_3 : \mathcal{S}_{\text{norm}} \times \mathcal{N} \rightarrow \mathcal{S}_{\text{enriched}} \quad (\text{Enrichment}) \quad (5)$$

$$\mathcal{F}_4 : \mathcal{N} \rightarrow \mathcal{C} \quad (\text{Critical Signals}) \quad (6)$$

$$\mathcal{F}_5 : \mathcal{S}_{\text{enriched}} \times \mathcal{C} \rightarrow \mathcal{G} \quad (\text{Branch Gating}) \quad (7)$$

$$\mathcal{F}_6 : \mathcal{G} \times \mathcal{S}_{\text{enriched}} \times \mathcal{C} \rightarrow \mathcal{R} \quad (\text{Scenario Gen.}) \quad (8)$$

$$\mathcal{F}_7 : \mathcal{R} \times \mathcal{N} \rightarrow \mathcal{E} \quad (\text{Attribution}) \quad (9)$$

C. Constrained Generation

Scenario generation $\mathcal{F}_6$ is subject to evidence constraints:

$$\forall r \in R_b : \exists e \in \mathcal{E} : \text{grounds}(r, e) = \text{true} \quad (10)$$

That is, every claim r in branch b 's scenario must be groundable in some evidence span e from the source note. Branches failing this constraint are flagged as “ungrounded.”

IV. PROPOSED METHODOLOGY

This section presents the architecture of the proposed framework, which transforms unstructured clinical notes into evidence-attributed crisis scenarios through a seven-stage pipeline. The stages are: (1) clinical factor extraction via LoRA fine-tuned language model, (2) post-extraction normalization, (3) context enrichment, (4) negation-aware critical signal detection, (5) evidence-based branch gating, (6) Tree-of-Thoughts scenario generation, and (7) evidence attribution with nexus factor detection. Fig. 1 illustrates the complete system architecture.

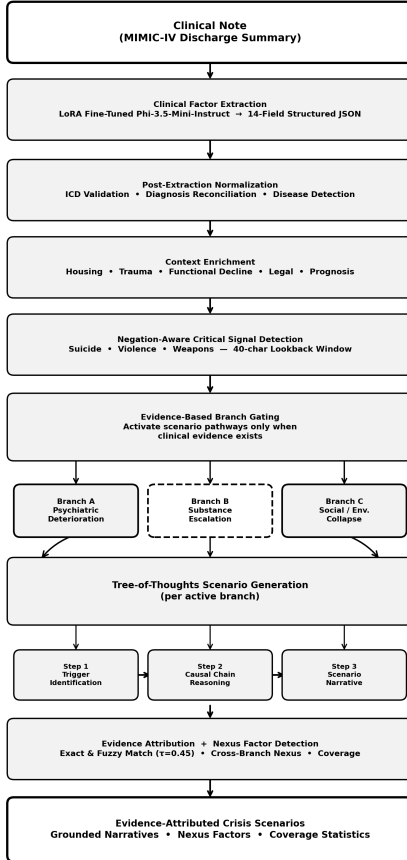


Fig. 1. System architecture: seven-stage pipeline from clinical note to evidence-attributed crisis scenarios.

Algorithm 1 presents the complete pipeline procedure. Given a clinical note $\mathcal{N}$, the pipeline produces structured clin-

ical factors $\mathcal{S}$, scenario narratives $\mathcal{R}$, and evidence attribution mappings $\mathcal{E}$.

Algorithm 1 Complete Pipeline Procedure

Require: Clinical note $\mathcal{N}$, fine-tuned model $\mathcal{M}$

Ensure: Factors $\mathcal{S}$, Scenarios $\mathcal{R}$, Attribution $\mathcal{E}$

```

1: Stage 1:  $\mathcal{S}_{\text{raw}} \leftarrow \text{Extract}(\mathcal{M}, \mathcal{N})$ 
2: Stage 2:  $\mathcal{S}_{\text{norm}} \leftarrow \text{Normalize}(\mathcal{S}_{\text{raw}}, \mathcal{N})$ 
3: Stage 3:  $\mathcal{S} \leftarrow \text{Enrich}(\mathcal{S}_{\text{norm}}, \mathcal{N})$ 
4: Stage 4:  $\mathcal{C} \leftarrow \text{DetectSignals}(\mathcal{N})$ 
5: Stage 5:  $\mathcal{G} \leftarrow \text{GateBranches}(\mathcal{S}, \mathcal{C})$ 
6: Stage 6:  $\mathcal{R} \leftarrow \emptyset$ 
7: for each branch  $b \in \{A, B, C\}$  do
8:   if  $\mathcal{G}[b] = \text{ACTIVE}$  then
9:      $\mathcal{R}[b] \leftarrow \text{TreeOfThoughts}(b, \mathcal{S}, \mathcal{N})$ 
10:  else
11:     $\mathcal{R}[b] \leftarrow \text{"NOT APPLICABLE"}$ 
12:  end if
13: end for
14: Stage 7:  $\mathcal{E} \leftarrow \text{AttributeEvidence}(\mathcal{R}, \mathcal{N})$ 
15: return  $(\mathcal{S}, \mathcal{R}, \mathcal{E})$ 

```

A. Stage 1: Clinical Factor Extraction

The extraction stage employs a fine-tuned large language model to transform unstructured discharge summaries into structured JSON representations conforming to a DSM-5/ICD-11 grounded schema.

Base Model. We employ Microsoft Phi-3.5-Mini-Instruct (3.82B parameters), a decoder-only transformer architecture comprising 32 layers with 32 attention heads, hidden dimension $d_{\text{model}} = 3072$, and vocabulary size $|\mathcal{V}| = 32,064$. Although the model supports 128K token context, we constrain sequences to 2048 tokens for computational efficiency.

LoRA Fine-Tuning. Low-Rank Adaptation (LoRA) [16] enables parameter-efficient fine-tuning by freezing pre-trained weights W_0 and learning low-rank decomposition matrices. For weight matrix $W_0 \in \mathbb{R}^{d \times k}$, the adapted weight is:

$$W = W_0 + \frac{\alpha}{r} BA \quad (11)$$

where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, rank $r = 32$, and scaling factor $\alpha = 64$. We apply LoRA to all attention and feed-forward projections: $\{W_q, W_k, W_v, W_o, W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}\}$, yielding 50.3M trainable parameters (1.30% of total).

Training Objective. The model minimizes cross-entropy loss over target JSON sequences:

$$\mathcal{L} = -\frac{1}{T} \sum_{t=1}^T \log P_{\theta}(y_t | y_{<t}, \mathcal{N}) \quad (12)$$

where $\mathcal{N}$ is the input note and $y = (y_1, \dots, y_T)$ is the target JSON.

Extraction Schema. The 14-field extraction schema is defined in Table I. Each and every field has constrained value spaces ensuring structured, validated output.

TABLE I
CLINICAL FACTOR EXTRACTION SCHEMA

Field	Description	Type
primary_dx	Primary diagnosis	ICD-10 F01-99
dsm5_category	DSM-5 classification	Categorical
severity	Symptom severity	4-level scale
suicide_risk	Ideation, plan, attempt	Boolean $\times$ 3
substance_use	Alcohol, drugs, toxicology	Multi-field
medications	Current prescriptions	List
social_support	Family, housing, employment	Multi-field
treatment_history	Prior hospitalizations	Integer

B. Stage 2: Post-Extraction Normalization

Raw extractions can have inconsistencies that have to be validated. This stage performs three normalization steps.

ICD Code Validation. ICD code extractions are checked against an authoritative lookup table $\mathcal{I}$ mapping codes to DSM-5 categories. Mismatched codes are fixed:

$$c' = \begin{cases} \mathcal{I}(c)_{\text{corrected}} & \text{if mismatch detected} \\ c & \text{otherwise} \end{cases} \quad (13)$$

Diagnosis Reconciliation. The primary diagnosis extracted is cross referenced to the section of pattern matched extraction of the primary diagnosis, which is the Discharge Diagnosis section, to verify consistency with clinical conclusions recorded.

Redaction Resolution. For privacy-redacted entities (e.g., “[REDACTED] Disease”), Contextual co-occurrence of keywords is used to infer the original diagnosis. An example is that the presence of “chorea” and “hereditary” implies with high confidence that it is Huntington disease.

C. Stage 3: Context Enrichment

The set extraction schema was not able to capture all clinically relevant factors. So the enrichment module searches the note for seven categories using pattern matching:

$$\mathcal{E}_{\text{context}} = \bigcup_{c \in \mathcal{C}} \text{PatternMatch}(\mathcal{N}, \mathcal{P}_c) \quad (14)$$

where $\mathcal{C}$ can include housing, caregiver, trauma, weight, functional, prognosis, and legal factors. Example patterns to be used: `homeless/evict*/shelter` for housing; `sexual*abus*/PTSD` for trauma.

D. Stage 4: Negation-Aware Critical Signal Detection

Clinical notes frequently document both active concerns and explicitly denied symptoms. The difference between “threatened to jump” (active signal) from “denies suicidal ideation” (negated signal) is the main key for accurate and proper risk assessment.

The detection procedure is formalised in Algorithm 2. A 40 character lookback window is scanned on each pattern match for negation terms. Negators (before the match: “denies,” “no,”

“negative for” etc.) are considered negated; the rest are considered active signals that need to be addressed immediately.

Algorithm 2 Negation-Aware Signal Detection

Require: Note $\mathcal{N}$, pattern sets $\mathcal{P} = \{\mathcal{P}_{\text{sui}}, \mathcal{P}_{\text{vio}}, \mathcal{P}_{\text{wpn}}\}$
Ensure: Active signals $\mathcal{A}$, Negated signals $\mathcal{D}$

```

1:  $\mathcal{A}, \mathcal{D} \leftarrow \emptyset, \emptyset$ 
2:  $\mathcal{N}_L \leftarrow \text{lowercase}(\mathcal{N})$ 
3:  $\text{NEG} \leftarrow \{\text{“denies”, “no”, “not”, “never”, “negative”}\}$ 
4: for each category  $c \in \{\text{sui, vio, wpn}\}$  do
5:   for each pattern  $p \in \mathcal{P}_c$  do
6:     for each match  $m$  at position  $i$  in  $\mathcal{N}_L$  do
7:        $\text{ctx} \leftarrow \mathcal{N}_L[\max(0, i - 40) : i]$ 
8:       if  $\exists n \in \text{NEG} : n \in \text{ctx}$  then
9:          $\mathcal{D} \leftarrow \mathcal{D} \cup \{(c, m, \text{“negated”})\}$ 
10:      else
11:         $\mathcal{A} \leftarrow \mathcal{A} \cup \{(c, m, \text{“active”})\}$ 
12:      end if
13:    end for
14:  end for
15: end for
16: return  $(\mathcal{A}, \mathcal{D})$ 
```

Categories of signals are like suicide indicators (“kill myself,” “end my life,” “overdose”), violence indicators (“assault,” “homicidal”), and weapons access (“gun,” “firearm”).

E. Stage 5: Evidence-Based Branch Gating

The framework uses conditional branch activation to avoid hallucinated scenarios which have no clinical basis. Three scenario branches are defined:

- **Branch A** (Psychiatric Deterioration): Always active
- **Branch B** (Substance Escalation): Conditional
- **Branch C** (Social/Environmental Collapse): Always active

Branch B activation requires evidence satisfying predicate ϕ_B :

$$\phi_B(\mathcal{S}) = (s_{\text{alcohol}} \neq \emptyset) \vee (s_{\text{drugs}} \neq \emptyset) \vee (s_{\text{tox}} = \text{positive}) \quad (15)$$

If $\neg\phi_B(\mathcal{S})$, Branch B outputs “NOT APPLICABLE No substance use evidence,” preventing fabricated substance abuse trajectories.

F. Stage 6: Tree-of-Thoughts Scenario Generation

For each active branch, structured counterfactual scenarios are generated through a three-step Tree-of-Thoughts (ToT) reasoning process. Algorithm 3 formalizes the procedure.

Step 1: Trigger Identification. The LLM identifies 3–5 triggering factors from the clinical note, each with direct evidence citations and mental health relevance ratings (scale 1–10).

Step 2: Causal Chain Reasoning. Given identified triggers, the LLM constructs 4–6 sequential escalation steps, each describing the mechanism linking the previous step to increased crisis risk.

Algorithm 3 Tree-of-Thoughts Scenario Generation

Require: Branch b , factors $\mathcal{S}$, note excerpt $\mathcal{N}_e$, LLM $\mathcal{M}$
Ensure: Scenario narrative R_b

```

1: // Step 1: Trigger Identification
2: prompt1 ← BuildPrompt( $b, \mathcal{S}, \mathcal{N}_e$ , “triggers”)
3:  $T_b \leftarrow \mathcal{M}(\text{prompt}_1)$  {3–5 triggers with evidence}
4: if  $|T_b| = 0$  then
5:   return “No triggering factors identified”
6: end if
7: // Step 2: Causal Chain Reasoning
8: prompt2 ← BuildPrompt( $b, T_b, \mathcal{S}$ , “causal”)
9:  $C_b \leftarrow \mathcal{M}(\text{prompt}_2)$  {4–6 escalation steps}
10: // Step 3: Scenario Narrative Synthesis
11: prompt3 ← BuildPrompt( $b, T_b, C_b, \mathcal{S}, \mathcal{N}_e$ )
12:  $R_b \leftarrow \mathcal{M}(\text{prompt}_3)$ 
13: return  $R_b$ 

```

Step 3: Scenario Narrative Synthesis. The final step synthesizes triggers and causal chains into a coherent crisis narrative, including: (a) detailed scenario description, (b) observable warning signs, (c) projected crisis endpoint, (d) clinical preparedness recommendations, and (e) plausibility rating.

G. Stage 7: Evidence Attribution

The final stage grounds generated scenarios in source text, providing post-hoc explainability. Algorithm 4 formalizes the evidence attribution procedure.

Span Location. For each evidence quote q of Span Location. In each evidence quote q generated by each scenario, the algorithm will first try the exact substring matching (case-insensitive). In case of no exact matching, a fuzzy matching based on the Ratcliff-Obershelp similarity measure will be applied:

$$\text{Sim}(q, s) = \frac{2 \cdot |\text{LCS}(q, s)|}{|q| + |s|} \quad (16)$$

where LCS denotes the longest common subsequence. Spans with similarity ≥ 0.45 are accepted.

Nexus Factor Detection. An evidence span if it is supporting multiple branches will contain a *nexus factor* which is a single clinical element driving multiple crisis pathways. Nexus factors are assigned with highest intervention priority, as addressing them can prevent multiple escalation trajectories simultaneously.

Coverage Statistics. Attribution coverage quantifies the proportion of the source note directly supporting generated scenarios:

$$\text{Coverage} = \frac{\sum_{e \in \mathcal{E}} |e_{\text{end}} - e_{\text{start}}|}{|\mathcal{N}|} \quad (17)$$

V. EXPERIMENTAL SETUP

A. Dataset: MIMIC-IV

We use MIMIC-IV v3.1 [12], filtering for mental health-related admissions:

$$\mathcal{D} = \{(\mathcal{N}_i, \mathcal{S}_i) : \text{ICD}_i \in [\text{F01–F99}] \cup [290–319]\} \quad (18)$$

Algorithm 4 Evidence Attribution with Nexus Detection

Require: Scenarios $\mathcal{R}$, note $\mathcal{N}$, threshold $\tau = 0.45$
Ensure: Attribution map $\mathcal{E}$, nexus factors $\mathcal{X}$

```

1:  $\mathcal{E} \leftarrow \{\}; \mathcal{X} \leftarrow \emptyset$ 
2: for each branch  $b \in \{A, B, C\}$  do
3:    $Q_b \leftarrow \text{ExtractQuotes}(\mathcal{R}[b])$ 
4:   for each quote  $q \in Q_b$  do
5:     // Try exact match first
6:     if  $q_{\text{lower}} \in \mathcal{N}_{\text{lower}}$  then
7:        $\mathcal{E}[b] \leftarrow \mathcal{E}[b] \cup \{(q, \text{span}, \text{“exact”})\}$ 
8:     else
9:       // Fall back to fuzzy match
10:       $s^* \leftarrow \arg \max_{s \in \text{Windows}(\mathcal{N})} \text{Sim}(q, s)$ 
11:      if  $\text{Sim}(q, s^*) \geq \tau$  then
12:         $\mathcal{E}[b] \leftarrow \mathcal{E}[b] \cup \{(q, s^*, \text{“fuzzy”})\}$ 
13:      end if
14:    end if
15:  end for
16: end for
17: // Detect nexus factors
18: for each span  $e$  across all  $\mathcal{E}[b]$  do
19:   if  $|\{b : e \in \mathcal{E}[b]\}| \geq 2$  then
20:      $\mathcal{X} \leftarrow \mathcal{X} \cup \{e\}$ 
21:   end if
22: end for
23: return  $(\mathcal{E}, \mathcal{X})$ 

```

TABLE II
DATASET COMPOSITION

Split	Patients	Examples
Training	45,223	94,592
Validation	5,640	11,409
Test	5,640	11,916
Total	56,503	117,917

Label Distribution: Severity levels (mild: 28.0%, moderate: 24.6%, severe: 34.9%, critical: 12.5%); DSM-5 categories (depressive: 31.9%, anxiety: 20.0%, substance: 19.5%, neurocognitive: 14.0%, other: 14.6%).

B. Training Configuration

VI. RESULTS AND ANALYSIS

This section includes the results of the experiment in three sections: (1) training of the extraction model, (2) development of the pipeline based on the initial system to the final one, and (3) qualitative evaluation that shows the capabilities of the system.

A. Model Training Results

The LoRA fine-tuned Phi-3.5-Mini-Instruct model was trained on 94,592 MIMIC-IV discharge summaries over 5 epochs (7,390 steps). Fig. 2 shows the training dynamics.

The final training metrics are summarized in Table IV. The model has a token-level accuracy of 67.5% which means that

TABLE III
HYPERPARAMETER CONFIGURATION

Parameter	Value
Base model	Phi-3.5-Mini-Instruct (3.82B)
LoRA rank r	32
LoRA alpha α	64
LoRA dropout	0.05
Learning rate	2×10^{-4}
Batch size (effective)	64
Gradient accumulation	8 steps
Epochs	5
Warmup ratio	0.03
Max sequence length	2048 tokens
Precision	FP16
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)
Hardware	4× Tesla V100-SXM2-32GB
Distributed	PyTorch DDP

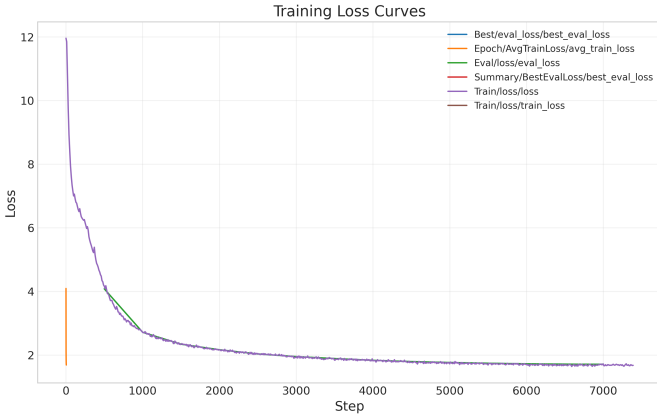


Fig. 2. Training and evaluation loss curves. Evaluation loss decreases from 4.08 to 1.706 (58.2% reduction) with no overfitting observed.

it can generate trustworthy JSON schemas. The perplexity of 5.51 indicates that the model is sure to make predictions of structured output tokens.

TABLE IV
TRAINING RESULTS SUMMARY

Metric	Value
Training time	35.8 hours
Total steps	7,390
Best evaluation loss	1.706
Final token accuracy	67.5%
Final perplexity	5.51
Trainable parameters	50.3M (1.30%)
Hardware	4× V100-32GB
Throughput	3.67 samples/sec
Total FLOPs	1.096×10^{19}

Table V presents the assessment measurements of every 500-step checkpoint. The model has a fast initial learning where it reaches an accuracy of 55.7% in the first 1,000 steps and then it is being refined slowly.

The gradient norms during training were also stable and

TABLE V
TRAINING PROGRESSION AT EVALUATION CHECKPOINTS

Step	Eval Loss	Token Acc.	Perplexity
500	4.079	41.6%	59.1
1,000	2.713	55.7%	15.1
2,000	2.164	61.8%	8.7
3,000	1.949	64.3%	7.0
4,000	1.832	65.8%	6.2
5,000	1.764	66.6%	5.8
6,000	1.722	67.2%	5.6
7,000	1.706	67.5%	5.5

there was a short initial spike to 2.97 at step 20, which rapidly stabilized to the 0.1-0.7 range thereafter. The warm-up schedule (222 steps) of 3% avoided the prolonged instability. The use of GPU memory was not much more than 11 GB per device (8.47 GB used, 10.13 GB max), which proves that LoRA can be efficiently fine-tuned using commodity hardware.

Fig. 3 shows the development of token accuracy. The accuracy grows quickly in the first epoch (steps 01478), with 55.7% accuracy at step 1000. Later epochs exhibit diminishing returns, as accuracy rises between 61.8% (step 2,000) and 67.5% (step 7,000) a trend that is consistent with the model learning structural JSON syntax early and perfecting semantic content extraction later epochs.

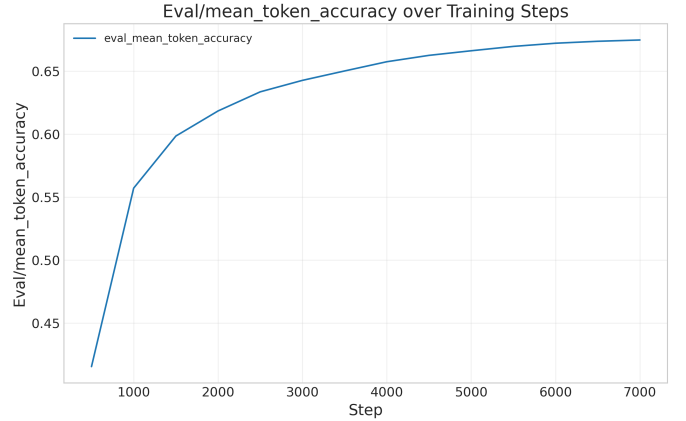


Fig. 3. Accuracy improvement at token level as a function of 7,390 training steps. Quick perfection in epoch 1 and gradual perfection. The last accuracy of 67.5% is a good indication of reliable JSON schema generation.

Fig. 4 is the perplexity trajectory. Perplexity values are used to quantify the uncertainty in prediction, lower values mean that the model puts more probability on correct tokens. The first perplexity of 59.1 (step 500) suggests that the model was just guessing between 59 equally likely tokens. Final perplexity of 5.51 is a 90.7% reduction, or the model is confident to reduce the predictions to about 5-6 candidate tokens on average.

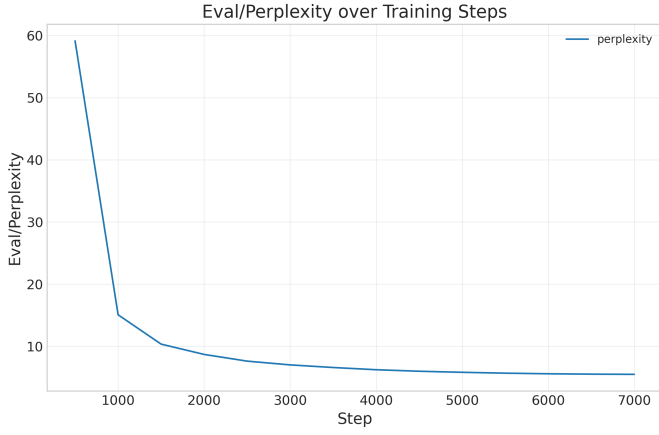


Fig. 4. Perplexity of evaluation over training. The perplexity was reduced by 59.1 to 5.51 (90.7% reduction), which means that the prediction confidence in the generation of structured output is much higher

B. Pipeline Evolution and Ablation Analysis

The framework was developed in four iterations where each iteration was in response to the shortcomings of the previous version. Table VI provides a summary of the development of the initial system to the final system.

TABLE VI
PIPELINE VERSION EVOLUTION

Ver.	Capabilities Added	Problems Addressed
V1	Extract → Scenarios → Report	Baseline pipeline
V2	+ICD normalization, +Critical signal scanner, +Branch gating, +Consistency checks	Extraction errors, missed safety signals, irrelevant scenarios
V3	+Context enrichment, +Negation awareness, +Deduplication	Missing social context, false positive alerts
V4	+Evidence attribution, +Nexus detection, +DOCX export	Lack of explainability, no traceability

1) *V1 Baseline: Three-Phase Pipeline*: The original system used a simple three-phase structure: (1) extraction of clinical factors using the fine-tuned model, (2) Tree-of-Thoughts scenario generation in three branches, and (3) synthesis of clinical reports. Although functional, V1 had a number of important limitations:

Problem 1: No validation layer. The errors in extraction were directly passed on to scenario generation. Misclassifications in ICD codes and improperly assigned severity remained in final productions.

Problem 2: Unconditional execution of branch. All the three branches of the scenarios (psychiatric, substance, social) were performed irrespective of their clinical relevance.

Fabricated substance abuse trajectories were given to patients who had no history of substance use.

Problem 3: No safety signal detection. The system was fully dependent on the LLM to detect important signals. In the case the model was not able to detect suicide or violence indicators, they were overlooked altogether.

Problem 4: Schema rigidity. The predetermined 14-field extraction schema overlooked contextually significant variables like housing instability, loss of caregivers, and trauma history.

2) *V2 Enhanced: Validation and Safety Pipeline*: Version 2 added four new modules to fill in the validation and safety gaps of V1:

Stage 1.5 (Normalization): Checks extracted ICD codes against an authoritative lookup table, reconciles primary diagnosis with the “Discharge Diagnosis” section, and categorizes diseases into DSM-5. This identified and fixed diagnostic errors prior to downstream processing.

Stage 1.6 (Critical Signal Detection): Applies rule-based regex scanning to suicide, violence, and weapons signals. This model-independent safety layer will have critical signals never missed, no matter how well the extraction model performs.

Stage 1.7 (Branch Gating): The stage assesses clinical evidence prior to the activation of each scenario branch. Branch B (Substance Escalation) needs documented alcohol use, drug use, or positive toxicology. Patients with no such evidence are labeled as “NOT APPLICABLE” instead of hallucinated substance scenarios.

Stage 3.5 (Consistency Checker): Does final validation with risk tier override when critical signals are identified, resolving the “unknown” severity values, and recognizing gated branches in the final report.

3) *V3 Refined: Context and Precision Pipeline*: Version 3 addressed the remaining gaps in contextual understanding and alert precision:

Stage 1.55 (Context Enrichment): Scans the clinical note for seven contextual categories not captured by the extraction schema: housing crisis, caregiver loss, trauma history, weight change, functional decline, clinician prognosis, and legal status. These enhanced factors are injected in the structured representation to generate downstream scenarios.

Negation-Aware Signal Detection: The critical signal scanner was improved with a 40-character lookback window to detect the negation terms. The sentence “Self-injury: denies” will now be properly interpreted as negated, whereas the sentence “threatened to jump out her window” will alert the system. This removed false positive alerts that afflicted V2.

Alert Deduplication: Alerts with the same signal pattern are merged before being emitted, reducing redundant alerts.

Table VII summarizes the measured error patterns across versions based on qualitative inspection of pipeline outputs during development.

4) *V4 Final: Explainable Attribution*: Although V3 generated clinically appropriate scenarios, it could not be explained in a way that could be verified by clinicians as to what parts of

TABLE VII
OBSERVED ERROR PATTERNS ACROSS PIPELINE VERSIONS

Error Type	V1	V2	V3
ICD code misclassification	Frequent	Rare	Rare
Missed critical signals	Frequent	None	None
Irrelevant substance scenarios	Frequent	None	None
False positive alerts	–	Frequent	None
Missing housing/trauma context	Frequent	Frequent	Rare

the original note underpinned each generated claim. Version 4 tackles this by:

Stage 4 (Evidence Attribution): Maps all scenario triggers and risk factors to particular spans of the original clinical note. The algorithm initially tries exact substring matching; failing which fuzzy matching using Ratcliff-Obershelp similarity ($\tau \geq 0.45$) finds evidence paraphrasing evidence supporting multiple branches at once.

Nexus Factor Detection: Finds evidence spans that support multiple branches at once. The fact that a psychiatric deterioration (Branch A) and a social collapse (Branch C) are mentioned as supporting a “housing crisis” is highlighted as a nexus factor with a high intervention priority.

Coverage Statistics: Calculates the percentage of the source note that is literally grounding generated scenarios. A greater coverage means enhanced evidence basis; low coverage indicates possible speculative outputs.

C. Illustrative Input-Output Example

We provide real-life examples of processed clinical note before presenting the entire qualitative case study to illustrate how the framework converts raw documentation into preparedness output in a structured format. Table VIII shows verbatim input text and corresponding system outputs.

This example shows that the framework can be used to achieve evidence-based transformations: language triggers enrichment and activation of Branch C, explicit threats result in critical alerts with risk growth, negated statements are suppressed accordingly, negative toxicology blocks the pathway, and functional indicators are summed up into enrichment factors in support of scenario plausibility. The above input and output are based on the MIMIC-IV discharge summary (subject_id: 11399055, hadm_id: 25710274).

D. Qualitative Case Study

On a representative instance, based on the MIMIC-IV test set (subject_id: 11399055, hadm_id: 25710274):

Patient Profile: Female with Huntington disease who presented with agitation after being informed of an apartment auction. Note length: 4,832 characters.

Stage 1 (Extraction): The fine-tuned model finds primary diagnosis (F02.80, Huntington disease with behavioral disturbance), severity (severe), and medication history. Fields of suicide risk: ideation=true, plan=false, attempt=false.

Stage 2 (Normalization): ICD code F02.80 validated against lookup table. Disease name originally redacted as

TABLE VIII
ACTUAL INPUT-OUTPUT EXAMPLE FROM PIPELINE

Input Note (Verbatim)	System Output
“Her apt has been auctioned off and will be homeless at the end of the week.”	[ENRICH] Housing crisis detected: homelessness, imminent homelessness, housing loss. Updated housing_status to ‘homeless’. Branch C activated.
“She threatened to jump out her 10 story window if she took her to one of her outpatient treators.”	CRITICAL SAFETY ALERT: SUICIDE/SELF-HARM SIGNAL: ‘jump out’. Risk tier override: HIGH.
“Self-injury: denies” / “Access to weapons: denies”	[NEGATED] violence: ‘weapons’ correctly classified as negated and excluded from active alerting.
Toxicology: “Ethanol-NEG, opiates-NEG, cocaine-NEG, amphetm-NEG”	Branch B: NOT APPLICABLE No substance use evidence. Data: alcohol=none, drugs=none, tox_screen=negative.
“wide-based and ataxic” / “monosyllabic phrases” / “difficulty caring for self”	[ENRICH] Functional decline detected: self-care difficulty, ataxic gait, monosyllabic speech, speech latency. Homebound duration: a year.

“[REDACTED] Disease” is resolved to “Huntington’s” via keyword co-occurrence (“chorea,” “hereditary,” “CAG repeat”).

Stage 3 (Enrichment): Seven enrichment categories evaluated:

- Housing crisis: DETECTED (“apartment auctioned,” “will be homeless”)
- Functional decline: DETECTED (“ataxic gait,” “monosyllabic speech”)
- Trauma history: DETECTED (“sexually molested in childhood”)
- Caregiver loss: DETECTED (“daughter refusing contact”)
- Prognosis: DETECTED (“progressive neurodegenerative”)

Stage 4 (Critical Signals):

- ACTIVE: “threatened to jump out her 10 story window”
- NEGATED: “Self-injury: denies” (correctly downgraded)
- NEGATED: “Access to weapons: denies” (correctly downgraded)

Stage 5 (Branch Gating):

- Branch A (Psychiatric): ACTIVE
- Branch B (Substance): GATED no alcohol, drugs, or positive toxicology
- Branch C (Social): ACTIVE

Stage 6 (Scenario Generation): Tree-of-Thoughts reasoning produces two scenarios. Branch A projects rapid cognitive decline with suicide risk escalation triggered by housing loss and family abandonment. Branch C projects complete social system collapse with homelessness and loss of psychiatric follow-up.

Stage 7 (Evidence Attribution):

- Total evidence spans located: 11
- Exact matches: 8 (72.7%)
- Fuzzy matches: 3 (27.3%)
- Nexus factors: 2 (“homelessness,” “daughter refusing contact”)
- Note coverage: 14.2%

The two nexus factors housing crisis and family estrangement support both Branch A and Branch C scenarios, indicating that intervention addressing these factors could mitigate multiple crisis pathways simultaneously.

VII. DISCUSSION

The results that we got supports two central claims of this work. To begin with, structured clinical factor extraction can be trained with reliable parameter-efficient adaptation: the model achieved an evaluation loss of 1.706, token accuracy of 67.5% and perplexity of 5.51 with only 1.30% of total model parameters trained. Second, more than extraction is needed in order to generate reliable clinical scenarios. The progression from V1 to V4 shows that the post-extraction normalization, context enrichment, safety-aware filtering, branch gating, and evidence attribution are not peripheral additions; they are necessary components for converting raw LLM outputs to clinically interpretable preparedness scenarios.

A. Interpretation of Findings

The training curves shows that the model learns the output structure in the initial stages and semantic fidelity in the later stages. The accuracy of the tokens increased quickly during the initial training phase, suggesting that the model quickly learned the JSON schema and the general extraction format. The following slower yet steady reduction in evaluation loss and perplexity suggests that there is slow progress in the accuracy of clinically meaningful field values and not just formatting. This difference is important because structurally valid output alone cannot be used in a clinical pipeline, downstream steps require semantically accurate diagnoses, severity labels, and risk indicators.

The version-wise pipeline analysis also gives us an equal significance result: scaling the base model alone did not give us the most effective system, but constraining and validating the behavior of the model step by step. V2 minimized lost critical signals and unnecessary execution of branches by adding normalization, explicit signal detection and branch gating. V3 also enhanced pragmatic reliability through the context enrichment and awareness of negations, which minimized false positive alerts and restored clinically significant social factors that the fixed extraction schema had ignored. V4 that we have built on the clinically plausible generation system to

explainable generation by tracing generated claims to spans of source evidence. Combined, these results that we got indicates that the architectural protections and post-hoc checking play a very important role in clinical NLP with high stakes as much in system trustworthiness as the underlying generative model.

B. Clinical and Methodological Implications

The mental health AI, in terms of clinical presentation, is moved to a prospective preparedness model as opposed to a retrospective risk classification. Instead of producing a scalar risk score, the system will find plausible deterioration pathways, the clinical and social triggers of those pathways, and the source-note evidence of the pathways that supports them. It is a more practical form of discharge planning, psychiatric triage, and crisis prevention as it facilitates intervention planning based on tangible elements such as housing instability, family estrangement, or active suicidality.

This change is depicted in the case study. The framework did not merely find a high-risk patient, it divided psychiatric and social degeneration pathways, blocked the unsupported substance pathway, and found nexus factors common to active branches. These outputs are more consistent with the reasoning of clinicians in practice regarding the trajectory of worsening, over which biological, psychiatric, and environmental stressors converge to escalate.

In terms of methodology, another practical design principle of domain-specific LLM systems is also noted: deterministic modules are useful even in the presence of a strong generative model. Rule-based negation detection, branch gating, ICD reconciliation, and evidence matching are examples of failures in generative models that are not consistently addressed by generative models. The resulting hybrid architecture is not thus a workaround, but a deliberate design decision towards better safety, controllability, and interpretability.

C. Future Work

A number of directions are naturally emerging out of these findings. The next step is the most immediate rigorous expert evaluation with manually annotated corpora to extract structured information, branch activation accuracy, and quality of evidence attribution. This type of evaluation would allow quantitative comparison with other clinical NLP baselines and would allow determining whether the perceived qualitative improvements of V3 and V4 have a quantifiable improvement in reliability.

In addition to annotation, clinician-in-the-loop experiments are required to determine whether generated scenarios can be helpful in the actual decision processes. Specifically, the research needs to be extended in future to investigate whether the use of scenario narratives, identification of nexus factors, and evidence-based explanations enhance preparedness planning, discharge decision making, or early intervention prioritization compared to the traditional risk summaries.

Longitudinal extensions at the modeling level is a promising direction. Multitasking with encounters, medication histories, past hospitalizations, and risk markers that change might be

useful in counterfactual simulation of trajectories, as opposed to note-level generation of scenarios. Equally, the evidence attribution might be enhanced through text matching and learned alignment techniques, grounding based on retrieval, or attention-informed attribution to optimally capture paraphrastic support.

Lastly, to determine strength, equity, and transportability, larger validation on psychiatric, emergency, and community mental health data will be required. This kind of work is necessary prior to the deployment-oriented claims.

VIII. CONCLUSION

This work addresses the main limitation of current mental health AI systems : Although previous approaches have the ability to classify risk, they do not explain how deterioration can occur or which documented factors support that trajectory. In order to address this gap, we have proposed a seven-stage framework for counterfactual simulation of extreme mental health scenarios from clinical notes. This framework combines LoRA-based clinical factor extraction, post-extraction normalization, context enrichment, negation-aware critical signal detection, evidence-based branch gating, Tree-of-Thoughts(TOT) scenario generation, and post-hoc evidence attribution.

The experimental results show that the extraction component can be learned effectively with parameter-efficient fine-tuning, achieving an evaluation loss of 1.706, token-level accuracy of 67.5%, and perplexity of 5.51 in 94,592 MIMIC-IV discharge summaries. Moreover, the version-wise evolution from V1 to V4 shows us that the reliable clinical scenario generation depends on the architectural safeguards, not just the base LLM. Normalization has reduced extraction inconsistencies, non-negation-aware signal detection has improved safety, branch gating prevented unsupported scenario generation, and evidence attribution has made the final outputs traceable to the source documentation.

The resultant system goes beyond black-box prediction toward evidence based clinical preparedness support. Rather than producing just a risk score, it identifies plausible crisis pathways, highlights shared nexus factors across pathways, and links generated claims back to supporting text spans in the note. This makes our framework more aligned with clinical reasoning and more suitable to be used in discharge planning, psychiatric triage, and early intervention workflows.

In general, this work contributes an end-to-end and explainable mental health crisis preparedness pipeline that integrates structured extraction, safety-aware validation, constrained generation, and traceable attribution within a single framework. While further clinician evaluation and external validation are necessary before deployment, the present results establishes a strong methodological foundation for evidence-based counterfactual reasoning in clinical mental health applications.

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